

A283 – Thermodynamics in FFGFT: a comprehensive overview

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Abstract. Thermodynamics is anchored in FFGFT at five levels — not as a postulate but as a consequence of geometry. This document collects the distributed individual results, arranges them into a unified chain of argument, and identifies where the levels interact and where they remain

strictly separate. A closing section concedes that both frameworks — expanding space and fractal inward refinement — are models that interpret the same observations in structurally opposite ways.

Layer markers. **[STIPULATION]** declared starting point — **[B]** algebraically/mathematically derived from declared foundations — **[K]** numerically computed from corpus constants — **[Q]** external primary source — **[S]** plausibility sketch — **[H]** open research question.

.1 Why a comprehensive document

The thermodynamic content of FFGFT emerged across several documents: Doc. 157 treats the arrow of time, A271–A282 the Landauer accounting, Doc. 313 the cosmological thermodynamics. Each document is self-contained; none provides an overview of how the parts relate to one another.

[B] Three questions remain unanswered without a comprehensive document: (1) Where do the levels touch, and where do they stay separate? (2) Which statements depend on the ξ scheme, which are independent? (3) What is the epistemic status of each statement? This document answers these questions and introduces no new claims.

.2 Level 1 — The arrow of time is geometric (Doc. 157)

The standard problem

All fundamental equations of physics are time-reversal invariant. The second law ($\Delta S \geq 0$) is statistical in nature and shifts the problem onto the unexplained low initial entropy (Penrose: $\sim 10^{-10^{123}}$).

The FFGFT answer

[B] Three geometric arguments interlock (Doc. 157):

Argument 1: Algebraic asymmetry. $\tilde{T} \cdot m = 1$ forces $\tilde{T} > 0$ because mass is positive definite. Time reversal would require $m \rightarrow -m$ — physically impossible.

Argument 2: Topological winding direction. The vacuum field winds with $\dot{\theta} = m = 1/\tilde{T}$ along the T^4 . This chirality is topologically protected.

Argument 3: Fractal dimensional asymmetry. $D_f = 3 - \xi_0 \approx 2.9999$ breaks the exact mirror symmetry; path integrals for forward and backward propagation differ by $O(\xi_0)$.

Consequence

[B] The logic is inverted: $\Delta S \geq 0$ in FFGFT is a *consequence* of the geometry, not a postulate. The “Past Hypothesis” becomes unnecessary — low initial entropy is geometrically forced, not an initial condition.

.3 Level 2 — Landauer as phase-space accounting (A271)

The phase-space formulation

[STIPULATION] A physical system occupies a region in phase space of size W (measured in units of h^f). The region exists independently of how one names or interprets it.

[B] A map that reduces several source regions $\{A, B\}$ onto a target region Z shrinks the volume: $\Delta S_{\text{system}} = -k_B \ln(W_{A+B}/W_Z) \leq 0$. Liouville's theorem forbids volume shrinkage in a closed system. A heat bath must absorb the volume:

$$Q \geq k_B T \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)$$

with $\ln 2$ as the binary special case.

Three separations

[B] (1) *Bit $\neq$ physical region*. Shannon and Boltzmann entropy coincide only when the description maps exactly onto the physical microstates with equal probability. (2) *Content neutrality*. Thermodynamics counts regions and does not read them. Bit "true" and bit "false" cost identically. (3) *Three levels strictly separate*. Construction / state transition / description. Landauer costs arise at levels 1 and 2, not at 3.

.4 Level 3 — Phase-space accounting universally (A280–A282)

[B] A280: The region comes first. A glass poured into the ocean — the volume does not vanish, it distributes over $1.4 \times 10^9 \text{ km}^3$. The price: the bath must absorb the volume, heat flows. The counting base is distinguishability (orthogonality), not energy; absolute base discretisation: h^f .

[B] A281: Landauer's own restriction — logically irreversible is a description property, not a physical one. Sufficient for heat is solely the non-bijectivity of the region map. Bennett's result: every logically irreversible function can be executed as a sequence of bijective region maps; costs attach to *forgetting*, not to the operation.

[B] A282: The token computer as a limiting case. The accounting holds exactly; the Landauer bound does *not* apply there (engineering costs dominate). The transition to the Brownian computing sphere connects continuously with Landauer's domain of applicability.

[H] Open question: minimal coherence cost for quantum coherence independent of error correction quality (A271, § 7.4).

.5 Level 4 — Cosmological thermodynamics (Docs. 312/313)

Thermal history as geometric condition

[K] Doc. 313: Three phases on the observable half-cycle ($\tau_c/2 \approx 46 \text{ Gyr}$):

Phase	Range	θ/π	Character
Radiation domination	$z > 3000$	> 0.90	$m \rightarrow m_{\min}$, slowest clocks
Transition	$z \approx 3000\text{--}1090$	$0.90\text{--}0.979$	recombination, scatter wall
Matter domination	$z < 3000$	< 0.90	$m(\theta)$ grows, structure formation

The five-step chain

[K] Complete derivation of Ω_m^* from ξ_0 alone:

Step	Result
(1) $H_0 = (\pi/2) c \xi_0^{10} / \bar{\lambda}_e$	$h = 0.6682$
(2) $k_B T_0 = (16/9) \xi_0$ (Doc. 061)	$T_0 = 2.751 \text{ K}$
(3) $\Omega_r \propto T_0^4 / H_0^2$	9.643×10^{-5}
(4) $K_{\text{frak}} = 1 - 100 \xi_0$ (A040)	0.98667
(5) $\int_0^\infty dz / E(z) = \pi / K_{\text{frak}}$	$\Omega_m^* = 0.3136$

Planck: $\Omega_m = 0.315 \pm 0.007$ (-0.19σ). Not a single step contains a parameter adjusted to Ω_m . Radiation ($\Omega_r \propto T_0^4 / H_0^2$) is the link between the particle and cosmic sectors.

Overdetermination and open core

[K] T_0 is determined via two independent routes: particle sector (Route 1: $k_B T_0 = (16/9) \xi_0$) and cosmic sector (Route 2: antipode condition). Both yield Ω_m agreeing to 0.1 % — the same overdetermination pattern as α in A130.

[K] Open (D(ii)): Coarse-grained entropy on the return half. Poincaré discrepancy: $\sim 10^{10^{104}}$ dynamical times against $\tau_c \approx 92 \text{ Gyr}$. Periodicity is a boundary condition, not a consequence of the dynamics. Three cases (A050/Doc. 295): Case A ($d = 0, 1$ cycle), Case B (ξ frozen, 75 cycles), Case C (ξ running, log-spiral — removes the entropy requirement, but costs τ_c).

Falsification points

[K] Low- ℓ cutoff: $\ell_{\min} = 2\pi D_{\text{LSS}} / L^* = 3.08$ — matches the quadrupole suppression unexplained since COBE (1992). Primary falsification test: $m_\tau = 1776.97 \text{ MeV}$ (Belle-II).

.6 Level 5 — What separates the levels

[B] The Landauer accounting (levels 2 and 3) does not touch the ξ scheme. It holds in any system with Hamiltonian or unitary dynamics — independent of the torus geometry. Explicitly booked in A271: “Contact with the ξ scheme of FFGFT: none.”

The arrow of time (level 1) is directly tied to $D_f = 3 - \xi_0$ and $\tilde{T} \cdot m = 1$ — FFGFT-specific. The cosmological thermodynamics (level 4) depends entirely on ξ_0 , H_0 and the torus geometry.

Level	Document	ξ dependence	Status
Arrow of time	Doc. 157	direct	[B]/[S]
Landauer	A271	none	[B]
Universal accounting	A280–A282	none	[B]
Cosmological thermo.	Docs. 312/313	direct	[K]
Information physics	Docs. 243–301	indirect	[S]/[B]

.7 Chain of argument: from geometry to heat

[B] The five levels form a chain from FFGFT geometry to classical thermodynamics: (1) ξ_0 and T^4 fix the geometry. (2) $\tilde{T} \cdot m = 1$ and the torus winding define a time direction geometrically. (3) In this directed time there is more phase-space volume in the forward direction — $\Delta S \geq 0$ as a geometric consequence. (4) Non-bijective phase-space maps (Landauer) cost heat — independently of ξ_0 , in the same phase-space picture. (5) The thermal history (Ω_m^* , Λ^*) follows from ξ_0 directly.

Liouville’s theorem and the second law are *consequences* in this chain, not postulates.

.8 Information thermodynamics in the IPI context (Docs. 243–301)

[S] A further group of documents connects FFGFT thermodynamics with the information-physics community (IPI context): Doc. 251 (comparison with Vopson’s information-mass hypothesis), Doc. 247 (Landauer’s principle in the context of information states), Doc. 290 (information question: magnitude/phase), Doc. 301 (information reversal test). These documents are plausibility sketches or partial algebraic derivations; they open connections to current debates without contributing new core results to FFGFT thermodynamics.

.9 Phases, radiation and temperature in the time cycle (Doc. 313 in detail)

A gradient on a circle

[B] The observable history shows a clear thermal sequence: radiation domination, transition, matter domination. This gives the impression of a directed process with a beginning. The FFGFT answer (Doc. 313): the gradient is not a development *from* a beginning but a map *along* the time cycle — like climate zones along a meridian circle.

Three phases on the half-cycle

[K] The observable history fills one half-cycle ($\tau_c/2 \approx 46$ Gyr). Cycle positions:

Event	z	θ/π	
		smooth	fractal
Matter-radiation equality	~ 3400	0.93	0.92
Scatter wall (recombination)	1090	0.992	0.979
Particle horizon	$z_{\max}$	1.012	0.9986

Phase 1 — Radiation domination (near the antipode)

In the mass-flow reading: masses $m(\theta)$ are minimal, intrinsic times $\tilde{T} = 1/m$ maximal — clocks run slowest. The endpoint slope of the mass profile at the antipode:

$$\left| \frac{dm}{d\theta} \right|_{\text{antipode}} = \sqrt{\Omega_r} \approx 0.0096$$

No free parameter — the observed radiation density appears here as the *smoothness condition* for periodic closure.

Phase 2 — Matter-radiation transition ($z \approx 3000$ to 1090)

With the fractal path correction $K_{\text{frak}} = 1 - 100\xi_0$ (A040, independently derived) the particle horizon sits at 0.14 % from the antipode — improvement by factor 9 over the smooth calculation (1.2 %), no adjusted parameter.

The path correction applies to path/structure, cancels for path/path. Invariance test: CMB peak positions unchanged (test passed without being designed for it).

Phase 3 — Matter domination (today)

Energy density $\propto (1+z)^3$. In the mass-flow reading masses grow with decreasing z ; clocks run faster; structure formation unfolds.

Temperature as a structural quantity

[K] In the Standard Model: $T(z) = T_0(1+z)$ — adiabatic back-extrapolation. In FFGFT: T_0 comes from the atomic scale (Doc. 061):

$$k_B T_0 = \frac{16}{9} \xi_0 \approx 2.370 \times 10^{-4} \text{ eV} \quad (\text{measured: } 2.349 \times 10^{-4} \text{ eV, } +0.92 \%)$$

The CMB temperature is a *structural quantity* of the ξ scheme, not an endpoint of a cooling history.

[K] Range limit (Doc. 313): equilibria at other locations on the time cycle are *not* derivable from the current one by extrapolation. Unaffected: the Ω_m^* chain (runs via ξ_0 , not via adiabatic back-calculation).

Overdetermination of T_0

[K] Two independent routes: Route 1 (particle sector): $k_B T_0 = (16/9)\xi_0$, no cosmological input. Route 2 (cosmic sector): antipode condition, no particle-physics input. Both yield Ω_m agreeing to 0.1 % (0.3136 vs. 0.3139) — the same pattern as α in A130. Lever: $dT_0/T_0/d\Omega_m/\Omega_m \approx -12$.

Lower bound and Hawking

[K] Energy quantum of the time cycle (Doc. 312): $E_{\min} = \hbar H_0 = 2.28 \times 10^{-52} \text{ J}$. Finite $z_{\max} = m_e/m_{\min} - 1 = 3.59 \times 10^{38}$. Singularity excluded: masses cannot approach zero.

[K] One KMS rule links Gibbons–Hawking, Unruh and Hawking: $T = \hbar/(k_B \tau)$ with τ the local time period. The compact time direction delivers T_{GH} without a horizon. The family ladder meets the cosmos at $r_s = R_H/2$ (exactly 0.500). Hawking evaporation for $M_\odot$: 56 orders of magnitude too slow for τ_c — sharpens the entropy obstacle D(ii).

Low- ℓ cutoff

[K] On a torus of side length L^* no modes with wavelength $> L^*$ exist:

$$\ell_{\min} = \frac{2\pi D_{\text{LSS}}}{L^*} = 3.08$$

No adjusted parameter. Matches the quadrupole suppression ($\ell = 2$, factor $\sim 1/6$), unexplained since COBE (1992) in Λ CDM. Order-of-magnitude statement [K|P20]; exact mode summation on the torus still to be done.

Reliable and unreliable statements

[K] Three classes (Doc. 313):

Class	Examples	Reliability
Counts	$ \text{Aut}(D4) = 1152$, kissing number 24	exact
Same-level ratios	$\alpha, m_\mu/m_e$, Koide Q	to 10^{-10}
With scale bridge	G, H_0 , absolute masses	anchor needed
Number/duration quantities	$\eta = n_B/n_\gamma$, window durations	not transferable

BBN observables (Y_p , D/H, ^7Li): not a pure same-level ratio — the same applies to ΛCDM .

.10 Mass formation, fractal refinement and the illusion of expansion

What is absent at the antipode: metric sharpness

[K] All physical lengths tied to mass scale as $1/m$: Compton wavelength $\bar{\lambda} = \hbar/(mc)$, Bohr radius $a_0 \propto 1/m_e$, intrinsic time $\tilde{T} = 1/m$. At the antipode ($m \rightarrow m_{\min}$) all run toward their maximum. Without localisable massive objects *metric sharpness* is absent.

[B] $m = 0$ is structurally excluded: from $\tilde{T} \cdot m = 1$ with $T \geq \tau_0$ it follows that there is no empty state, only a thinnest one (A060). Mass and temporal curvature are two readings of the same structure — neither produces the other.

[K] The quantification follows from the double structure of length bounds. FFGFT has two cutoffs that follow from ξ alone:

- Lower length bound: $L_0 = \xi \ell_P$ (A060/A015) $\Rightarrow$ upper energy bound $E_{\max} = E_P/\xi$ (UV cutoff).
- Upper length bound: $L^* = 4\bar{\lambda}_e/\xi_0^{10}$ (Doc. 312) $\Rightarrow$ lower energy bound $E_{\min} = \hbar H_0$ (IR cutoff).

From $\tilde{T} \cdot m = 1$ and the IR bound:

$$m_{\min} = \frac{\hbar H_0}{c^2} = 2.54 \times 10^{-69} \text{ kg}$$

This is no additional stipulation — it is the cosmological side of the same reciprocity $\tilde{T} \cdot m = 1$ that A060 describes on the particle scale. A060 calls this range “unquantified” because the cosmological scale lies outside its scope (A010); Docs. 312/313 quantify it. The metric limiting case is thus fully determined: $m_{\min}$ is a derived value, not an open point.

Photons: no freezing

[B] Photons carry no rest mass — $\tilde{T} \cdot m = 1$ does not apply to them directly. A photon at the antipode does not freeze; it propagates at c . CMB photons lose energy geometrically on the way to us (A265):

$$\frac{dE}{dx} = -\xi E \Rightarrow 1 + z = e^{\xi x}$$

Achromatic: all wavelengths stretched by the same factor. From our perspective everything proceeds uniformly. The view back is a view to a *location* on the time cycle, not into the past.

Expansion as fractal inward refinement

[B] In the Standard Model: space expands outward, scale factor $a(t)$ grows. In FFGFT there is no expanding space. As $m(\theta)$ increases, all characteristic lengths ($\propto 1/m$) become *smaller*.

Matter becomes more structured, more finely resolved. This is **inward refinement**, not outward growth.

[B] $K_{\text{frak}} = 1 - 100\xi_0$ (A040): dimension deficit $D_f = 3 - \xi_0$, locally $\sim 0.003\%$, cumulatively over 17 orders of magnitude ~ 100 .

[B] Recursion $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ (A050), fixed point: zero. Rolled up: winding closes after 75 cycles (τ_c -cycle). Unrolled: logarithmic spiral $\rho = 75 e^{\beta/2\pi}$ — what the Standard Model calls expansion.

[K] Both descriptions are observationally equivalent (A265). Rulers and clocks change in the same ratio; no local observer detects the scaling. What we perceive as expansion is the contrast between our current masses and the redshifted light from the antipode.

[S] Image: Mandelbrot zoom. Zooming inward reveals ever finer structures, no outward growth. The antipode is the unzoomed view (minimal differentiation, pure radiation). Our current position is the deep-zoomed view (sharp atomic and stellar structures). The Standard Model inverts the picture — a choice of coordinates, not a physical statement.

.11 Open questions

[H] Three questions remain open: (1) Minimal coherence cost for quantum coherence independent of error correction quality (A271, §7.4). (2) Continuous entropy flux as a function of clock rate, bit width and logical irreversibility (A271, §8). (3) Metric structure in the limit $m \rightarrow m_{\min}$ — whether a regime of pure field geometry without localisable particle modes exists.

.12 Layer-status summary

Statement	Status
$\tilde{T} > 0$ forces geometric arrow of time	[B]
Torus winding: topologically protected direction	[B]
$D_f = 3 - \xi_0$ breaks time-reversal symmetry	[B]
$\Delta S \geq 0$ as geometric consequence	[B]
Landauer: $Q \geq k_B T \ln 2$ from Liouville	[B]
Content neutrality of thermodynamics	[B]
Bennett's reversible logic	[B]
$\Omega_m^* = 0.3136$ from ξ_0 chain	[K]
$\Lambda^* = 1/R_H^2$ geometric carrier	[K]
Overdetermination T_0 : two sectors to 0.1 %	[K]
Singularity excluded by $\hbar H_0$	[K]
$\ell_{\min} = 3.08$ (low- ℓ cutoff)	[K]P20
Slowest clocks at the antipode	[K]
Compton wavelength, Bohr radius $\propto 1/m$	[B]
Photons achromatically redshifted: $1 + z = e^{\xi x}$	[B]
Expansion = fractal inward refinement	[B]
Recursion: fixed point zero, spiral unrolled	[B]
Observational equivalence with Λ CDM	[K]
Landauer does not touch ξ scheme	[B]
Coarse-grained closure D(ii): open	open
$m = 0$ structurally excluded: no empty state (A060)	[B]
Double cutoff: $L_0 \Rightarrow E_{\max}, L^* \Rightarrow E_{\min}$	[K]
$m_{\min} = \hbar H_0 / c^2$ from IR cutoff (Docs. 312/313)	[K]
Local field modes resolvable at thinnest point?	[H]
Past Hypothesis as geometric necessity	[S]

Statement	Status
Mandelbrot zoom as image	[S]

Balance: 14× [B] · 8× [K] · 3× [S] · 2× [H] · 1× open.

.13 Closing remark: two opposing models

[S] An honest concession belongs at the end.

For us — as observers with our current masses and clocks — our perspective naturally prevails. We think in terms of light travel distances. We measure redshifts. We experience time as running forward. This is real in the sense that it is what we measure, and our instruments are calibrated to it.

The Standard Model (Λ CDM) takes exactly this perspective as its starting point: expanding space, the Big Bang as a beginning, redshift as spatial stretching, the CMB as a snapshot of an early epoch. It is a model in excellent agreement with observations.

FFGFT proposes a different choice of coordinates: no expanding space, no Big Bang as an event, redshift as geometric energy loss, the CMB as a view to a location on the time cycle. This description also agrees with the standard tests (A265, cosmological degeneracy).

[B] What distinguishes the two models:

- Λ CDM generalises from the observer's perspective *outward*: expanding space.
- FFGFT generalises from the geometry *inward*: compact torus, fractal refinement.

Both are models. Both step beyond direct observation and add an interpretive layer.

[S] What remains: two structurally opposite ways to describe the same thing. In one everything expands outward, in the other everything refines inward. In one time has a beginning, in the other it is a closed cycle. In one the CMB is a window into the past, in the other a window onto a location.

This is not a flaw — it is an open question, stated precisely enough to be decidable. The falsification tests (m_τ , $\ell_{\min}$, Ω_m^* chain) will show whether the geometric inward perspective of FFGFT carries more than the expanding outward perspective — or less.

.14 Source references

Doc. 061 Pascher, J., *Temperature units and CMB in T0 theory*. Legacy corpus, 2025.

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.15 Use of AI tools

This document was produced with the assistance of AI tools. The questions, stipulations and all substantive decisions originate with the author; the tools formulate and organise. Responsibility for content and errors rests entirely with the author. Full statement in A010.